Scalability & Future Plan

Date	23 October 2026
Team ID	[Your Team ID]
Project Name	TravelMate
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	AI output may contain inaccurate opening hours, travel times or prices.	Users may rely on wrong estimates.	High
2	Geocoding relies on Nominatim, which needs valid place names and has rate limits.	Some map pins can be missing.	Medium
3	The app depends on external services (Groq, Supabase, Render, map services).	An outage affects related features.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single FastAPI backend instance on Render.	Run more instances and queue AI requests.
2	Data Storage	Trips stored in Supabase.	Add indexes and archive old trips.
3	Performance	Client-side geocode cache and rate limiting.	Server-side caching of geocoding and AI results.
4	Security	Supabase Auth, CORS restricted to the frontend origin.	Audit RLS policies, add API rate limiting and rotate keys.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Weather information and trip sharing links	Next 1-2 months	More useful, shareable itineraries
Phase 3	Multi-currency budgets and collaborative trip editing	Next 3-4 months	Better support for groups and international trips
Phase 4	Mobile app / PWA and offline itinerary access	Next 6 months	Use of the plan while travelling